

Predictors of COVID-19 Protective Behavior Before and After Mask Mandates: A Cross-National Analysis

Zixuan Zhao A1948883

2026-6-23

1. Project background

- During **COVID-19**, protective behaviours such as **mask wearing**, social distancing, hand hygiene, and self-isolation were important for reducing transmission.
- Compliance was shaped not only by policy rules, but also by risk perception, trust, demographic characteristics, wellbeing, and broader social context.
- These predictors may operate **differently across countries** because policy environments, institutions, culture, and public expectations vary.

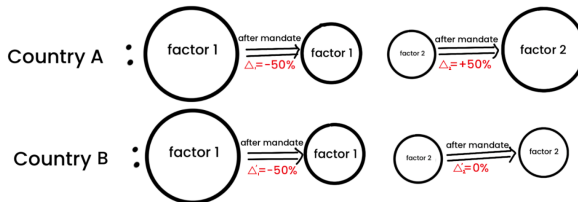
2. Why a cross-national comparison?

- Some studies suggest that certain predictors remain significant across different periods, while others only become significant before or after mandatory regulations are introduced.
- Evidence from a single country is not sufficient, because predictors that are important in one context may be less relevant in another.

3. Research questions

- Within each country, do the key predictors of compliance with COVID-19 protective behaviours change before and after mask mandates?
- Are these changes consistent across countries, or do they vary?
- Is there a set of common cross-national predictors, alongside country-specific predictors?

The influence of various factors on mask-wearing behaviour



Which factors show similar patterns of change before and after mandate across different countries ?
Which factors show different patterns of change in this regard?

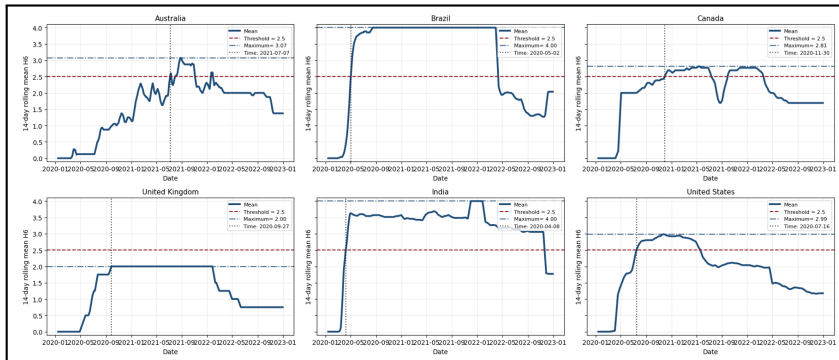
1. Behaviour and policy datasets

- The first dataset is the **Imperial College London / YouGov COVID-19 Behaviour Tracker**. It contains repeated cross-sectional survey data on protective health behaviours, demographics, risk perceptions, trust, wellbeing, and related variables.
- The second dataset is the **Oxford COVID-19 Government Response Tracker (OxCGRT)**. This records the strictness with which the government has enforced its mask-wearing policy.

2. How the datasets are linked

- Survey responses are linked to policy data by **country, region and date**.
- The mask mandate period is defined using the OxCGRT facial-covering policy indicator. When the stringency of a policy reaches a certain level, it is considered that the region has implemented a mandatory mask-wearing policy.

3. Policy timing figure



This figure show how mask policy intensity changed over time, and how the mandate threshold was used to define the before and after mandate periods.

4. Country coverage

The current cleaned cross-national sample includes six countries:

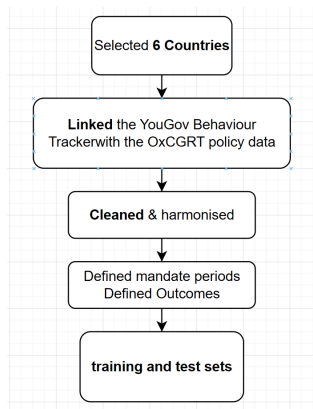
Country code	Country name
AUS	Australia
BRA	Brazil
CAN	Canada
CHN	China
GBR	United Kingdom
IND	India
USA	United States

China was excluded from the current analysis because several key variables needed for the harmonised workflow were not available.

What Have I Done So Far?

1. Data pre-processing

- Standardised raw YouGov and OxCGRT files across countries.
- Mapped survey regions to policy regions for correct linkage.
- Defined before/after mandate periods using the facial-covering policy indicator.
- Constructed two binary outcomes: face mask wearing and overall protective behaviour.
- Cleaned the data: Removing columns and samples with too many missing values, encoding the data, making column names more readable, and splits.



What Have I Done So Far?

2. About Model

1. Cleaned sample size

The preprocessing pipeline produced cleaned before- and after-mandate datasets for all six countries.

Country	Before mandate	After mandate	Total
Australia	15,757	25,988	41,745
Brazil	1,112	9,006	10,118
Canada	15,944	23,967	39,911
United Kingdom	7,838	24,580	32,418
India	1,736	13,629	15,365
United States	1,787	15,428	17,215

This shows that the pipeline has produced analysable data for each country and for both policy periods.

2. Train-test split

The next step was to prepare data for modelling.

- I split the data separately for each country, policy period, and outcome.
- I used an 80/20 train-test split.
- Stratification was used where possible to preserve the class balance of the outcome.

Dimension	Values	Number
Countries	AUS, BRA, CAN, GBR, IND, USA	6
Policy periods	Before mandate, after mandate	2
Outcomes	Face mask, overall protective behaviour	2
Total modelling tasks	$6 \times 2 \times 2$	24

3. Model Performance

Summary of current progres

- XXX
- XXX
- XXX

Next Step

- XXX

All data and code

All code can be found in my GitHub: <https://github.com/ZixaunZhao/COVID-19-Protective-Behavior-Cross-National-Analysis.git>